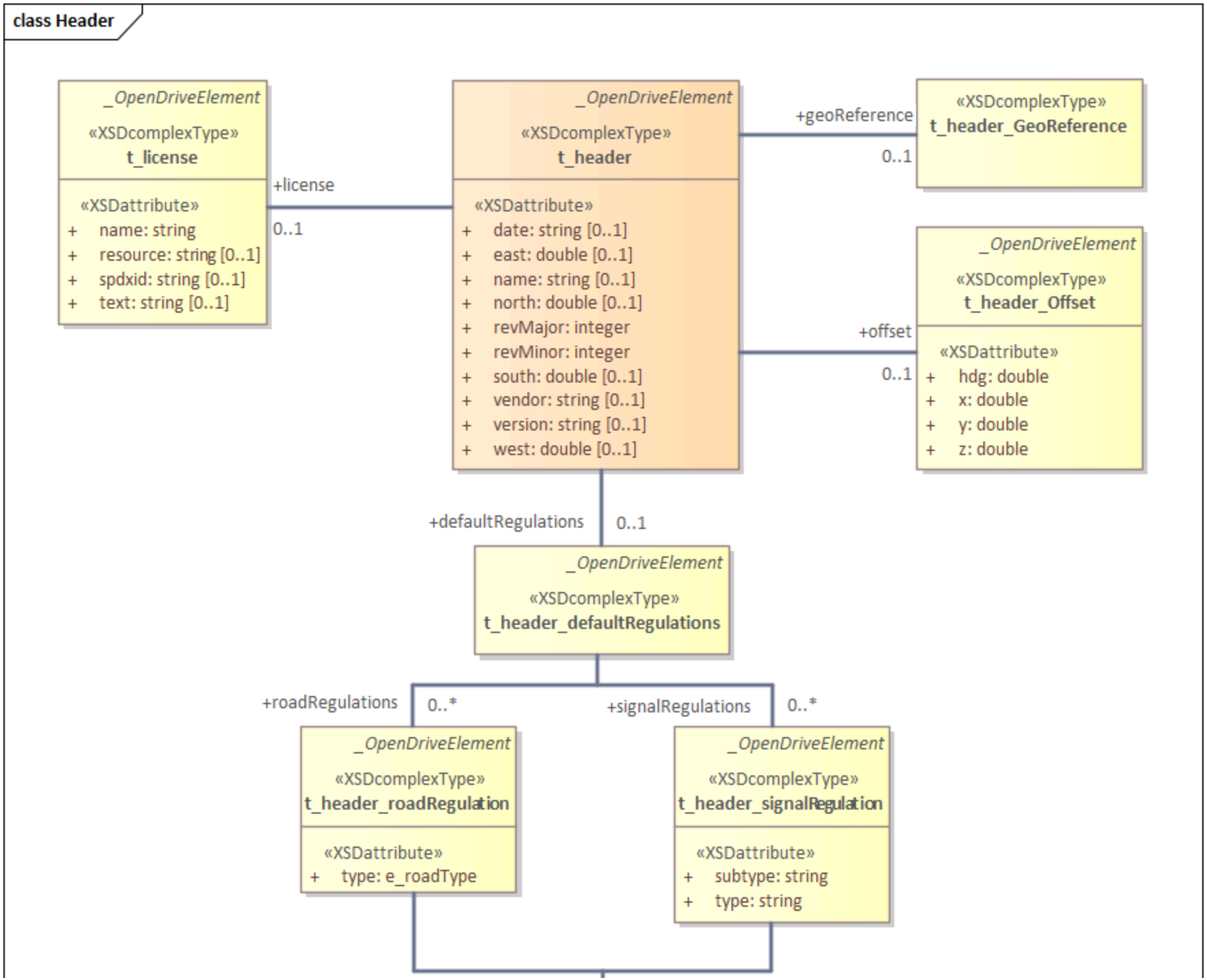
OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

6.4 Header

6.4.1. `<header>` element

The header is the very first element within the `<OpenDRIVE>` element. It is a mandatory element and contains general information about the ASAM OpenDRIVE file, for example version of the ASAM OpenDRIVE or default regulations.


class Header
_OpenDriveElement
«XSDcomplexType»
t_license
«XSDattribute»
+ name: string
+ resource: string [0..1]
+ spdxid: string [0..1]
+ text: string [0..1]
+license
0..1
_OpenDriveElement
«XSDcomplexType»
t_header
«XSDattribute»
+ date: string [0..1]
+ east: double [0..1]
+ name: string [0..1]
+ north: double [0..1]
+ revMajor: integer
+ revMinor: integer
+ south: double [0..1]
+ vendor: string [0..1]
+ version: string [0..1]
+ west: double [0..1]
+geoReference
0..1
«XSDcomplexType»
t_header_GeoReference
+offset
0..1
_OpenDriveElement
«XSDcomplexType»
t_header_Offset
«XSDattribute»
+ hdg: double
+ x: double
+ y: double
+ z: double
+defaultRegulations
0..1
_OpenDriveElement
«XSDcomplexType»
t_header_defaultRegulations
+roadRegulations
0..*
+signalRegulations
0..*
_OpenDriveElement
«XSDcomplexType»
t_header_roadRegulation
«XSDattribute»
+ type: e_roadType
_OpenDriveElement
«XSDcomplexType»
t_header_signalRegulation
«XSDattribute»
+ subtype: string
+ type: string

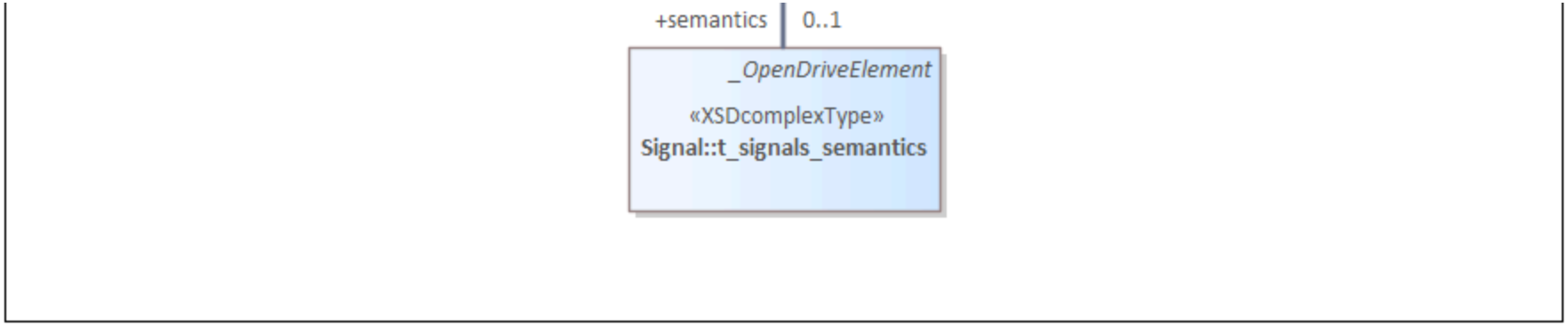


Figure 5. UML class diagram of the Header class

Figure 5 shows the UML class diagram of the ASAM OpenDRIVE Header class.

In ASAM OpenDRIVE, the general information is represented by the `<header>` element.

```
UML class: t_header
XML tag:   <header>
```

Contains general information about the ASAM OpenDRIVE file

Table 8. Attributes of the <header> element

Name	Type	Use	Unit	Description
`date`	string	optional		Time/date of database creation according to ISO 8601 (preference: YYYY-MM-DDThh:mm:ss)
`east`	double	optional	m	Maximum inertial x value
`name`	string	optional		Database name
`north`	double	optional	m	Maximum inertial y value
`revMajor`	integer	required		Major revision number of OpenDRIVE format

Name	Type	Use	Unit	Description
revMinor	integer	required		Minor revision number of OpenDRIVE format; 6 for OpenDrive 1.6
south	double	optional	m	Minimum inertial y value
vendor	string	optional		Vendor name
version	string	optional		Version of this road network
west	double	optional	m	Minimum inertial x value

NOTE

The `<geoReference>` and `<offset>` elements are explained in Section 8.5, "Georeferencing".

6.4.2. `<license>` element

```
UML class:  t_license
XML tag:    <license> (Multiplicity: 0..1)
Introduced: 1.8.0
```

Licensing information about the OpenDRIVE file.

Table 9. Attributes of the <license> element

Name	Type	Use	Introduced	Description
name	string	required	1.8.0	The full name of the license. Informational only.
resource	string	optional	1.8.0	Link to an URL where the full license text can be found.
spdxid	string	optional	1.8.0	The identifier of the license from the SPDX license list. Can also be an SPDX License Expression, which is also applicable to custom licenses (LicenseRef-...).

Name	Type	Use	Introduced	Description
text	string	optional	1.8.0	The full license text.

6.4.3. <defaultRegulations> element

```
UML class:  t_header_defaultRegulations
XML tag:    <defaultRegulations> (Multiplicity: 0..1)
Introduced: 1.8.0
```

Defines the default regulations. In each country there are different speed limits to a rural road. For example a rural road has a speed limit of 100km/h in Germany and 80km/h in the Netherlands.

In some countries, one is allowed to turn right at a red traffic light; in others, one is not. Instead of writing this for each road or each signal, the default regulations can be specified once in the header for the entire ASAM OpenDRIVE file. The default driving regulations can be over-written with road, lane, or signal definitions.

<roadRegulations> element

```
UML class:  t_header_roadRegulation
XML tag:    <roadRegulations> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Defines the default regulations for different road types.

Table 10. Attributes of the <roadRegulations> element

Name	Type	Use	Introduced
type	e_roadType	required	1.8.0

`<signalRegulations>` element

```
UML class:  t_header_signalRegulation
XML tag:    <signalRegulations> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Defines the default regulations for signs in different countries, for example, if it is allowed to turn right when a red traffic light appears.

Table 11. Attributes of the <signalRegulations> element

Name	Type	Use	Introduced
`subtype`	string	required	1.8.0
`type`	string	required	1.8.0

NOTE

The `<semantics>` element is explained in Section 14.8, "Signal semantics".

XML example

```xml
<OpenDRIVE>
    <header revMajor="1"
            revMinor="5"
            name=""
            version="1.00"
            date="Mon Nov 29 12:59:50 2021"
            north="0.0000000000000000e+00"
            south="0.0000000000000000e+00"
            east="0.0000000000000000e+00"
            west="0.0000000000000000e+00">
        <defaultRegulations>
            <roadRegulations type="motorway">
                <semantics>
                    <speed type="maximum" value="120" unit="km/h"/>
                </semantics>
            </roadRegulations>
            <roadRegulations type="rural">
                <semantics>
                    <speed type="maximum" value="50" unit="km/h"/>
                </semantics>
            </roadRegulations>
            <roadRegulations type="town">
                <semantics>
                    <speed type="maximum" value="30" unit="km/h"/>
                </semantics>
            </roadRegulations>
            <roadRegulations type="livingStreet">
                <semantics>
                    <speed type="maximum" value="5" unit="km/h"/>
                </semantics>
            </roadRegulations>
            <signalRegulations type="1000001" subType="-1">
                <semantics>
                    <priority type="turnOnRedAllowed" />
                </semantics>
            </signalRegulations>
```

```
        </defaultRegulations>
    </header>
</OpenDRIVE>
```

Rules

- asam.net:xodr:1.8.0:defaultRegulations.only_speed_prioity: Only `<speed>` and `<priority>` elements shall be used within the `<defaultRegulations>` element.
- The `<signalRegulations>` element shall be used for signs that result in different driving rules.
- `<signalRegulations>` elements shall not be used as a signal catalogue.